

Machine Learning Algorithms Laboratory

Experiment No.	3
Experiment Title	Regression Analysis using Linear and Regularized Models
GitHub Repository	https://github.com/Ray-Audrey/ML-Experiments
Student Name	Harshita S
Register Number	3122247001020
Faculty	Dr. Poreddy Ajay Kumar Reddy
Submission Date	27/07/2026

1 Objective

The objective of this experiment is to implement and analyze regression models for predicting loan sanction amounts using Linear Regression and regularization-based regression techniques.

The following regression algorithms are implemented and compared:

- Linear Regression
- Ridge Regression
- Lasso Regression
- Elastic Net Regression

The experiment also focuses on:

- Performing exploratory data analysis.
- Handling missing values and categorical features.
- Applying feature scaling and encoding techniques.
- Performing hyperparameter tuning using GridSearchCV.
- Comparing models using regression evaluation metrics.
- Studying the effect of regularization on model coefficients.

2 Problem Statement

Loan approval systems require accurate prediction of the sanctioned loan amount based on customer information, financial status, credit history, and property details.

The objective of this experiment is to develop machine learning regression models that can predict the loan sanction amount for a customer.

Input:

The input consists of customer-related features including:

- Age
- Income
- Credit Score
- Loan Amount Request
- Property Price
- Employment details
- Property information

Output:

The output variable is:

Loan Sanction Amount (USD)

The performance of different regression models is evaluated using:

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)
- R^2 Score

3 Dataset Description

The dataset contains customer financial information used for predicting loan sanction amounts.

Dataset Attribute	Description
Dataset Name	Loan Prediction Dataset
Dataset Source	Kaggle Loan Amount Prediction Dataset https://www.kaggle.com/datasets/phileinsophos/predict-loan-amount-data
Number of Samples	30000 samples
Number of Features	23 input features and 1 target variable
Target Variable	Loan Sanction Amount (USD)
Regression Type	Continuous value prediction
Missing Values	Present in numerical and categorical attributes
Train-Test Split	80% Training and 20% Testing
Training Samples	23457 samples
Testing Samples	5865 samples

The dataset contains numerical and categorical attributes.

Numerical features include:

- Age
- Income
- Loan Amount Request
- Credit Score
- Property Price
- Property Age

Categorical features include:

- Gender
- Profession
- Employment Type
- Location
- Credit Card Status
- Property Location

4 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to understand the structure, distribution, relationships, and quality of the dataset before model training.

The following analysis was carried out:

- Dataset shape and statistical summary.
- Missing value analysis.
- Duplicate record detection.
- Target variable distribution analysis.
- Feature-target relationship analysis.
- Correlation analysis between numerical features.

4.1 Dataset Overview

The dataset contains 30000 customer records with 24 attributes. Among these, 23 attributes are used as input features and Loan Sanction Amount (USD) is considered as the target variable.

The dataset contains both numerical and categorical attributes. Missing values were observed in several columns such as:

- Income (USD)

- Income Stability
- Type of Employment
- Credit Score
- Property Age
- Loan Sanction Amount (USD)

No duplicate records were present in the dataset.

4.2 EDA Summary Visualization

A consolidated EDA figure containing multiple subplots was generated to analyze the dataset characteristics.

The figure contains:

1. Target variable distribution
2. Age distribution
3. Income distribution
4. Credit score distribution
5. Loan request distribution
6. Property price distribution
7. Missing value analysis
8. Feature correlation heatmap
9. Loan amount request vs target
10. Property price vs target
11. Credit score vs target
12. Income vs target

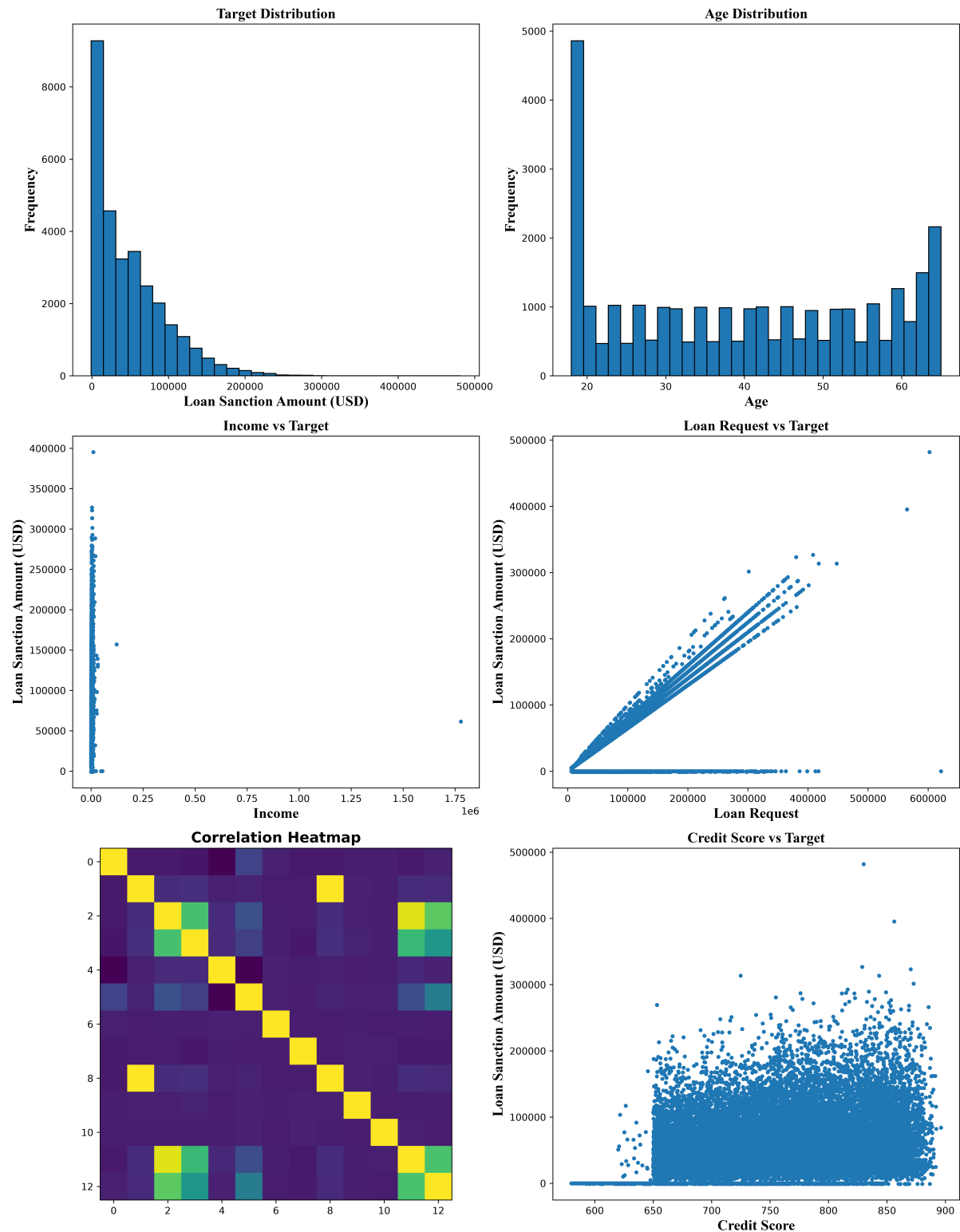


Figure 1: Consolidated Exploratory Data Analysis Visualization

Inference:

The EDA visualization provides an overview of the dataset distribution and feature relationships. The target variable Loan Sanction Amount shows a wide range of values with some variation. Loan Amount Request and Property Price show strong positive relationships with the target variable. Credit Score also shows a positive correlation

indicating its importance in loan sanction prediction. Missing value analysis highlights the need for proper pre-processing before model training.

4.3 Correlation Analysis

The correlation matrix was generated using numerical features to identify relationships between variables.

The highest correlations with the target variable were:

Feature	Correlation with Target
Loan Amount Request (USD)	0.7264
Property Price	0.6871
Current Loan Expenses (USD)	0.4851
Credit Score	0.3694
Income (USD)	0.0381

Table 2: Correlation of Features with Loan Sanction Amount

Inference:

Loan Amount Request and Property Price have the strongest influence on the loan sanction amount. Credit Score also contributes significantly because customers with better credit history generally receive higher sanctioned amounts. Other attributes such as Age and Dependents show weaker linear relationships with the target.

5 Data Preprocessing

Data preprocessing was performed to convert the raw dataset into a suitable format for regression algorithms.

5.1 Removal of Unnecessary Attributes

The following columns were removed because they do not contribute meaningful information for prediction:

- Customer ID
- Name
- Property ID

5.2 Missing Value Handling

Missing values were handled using suitable imputation techniques.

For numerical features:

- Missing values were replaced using mean imputation to reduce the effect of outliers.

For categorical features:

- Missing values were replaced using the most frequent category.

5.3 Handling Missing Symbols

The following symbols representing missing data were replaced with NaN:

- ?
- -999
- Empty values
- NA

5.4 Categorical Feature Encoding

Categorical variables were converted into numerical form using One Hot Encoding.

The encoding method creates separate binary features for each category while handling unknown categories during testing.

5.5 Feature Scaling

Numerical features were standardized using StandardScaler.

The transformation is:

$$X_{scaled} = \frac{X - \mu}{\sigma}$$

where:

- μ represents the mean of the feature.
- σ represents the standard deviation.

Feature scaling improves convergence and ensures that features with larger values do not dominate the regression models.

5.6 Train-Test Split

The processed dataset was divided into training and testing sets.

Dataset	Samples
Training Data	23457
Testing Data	5865

Table 3: Train-Test Dataset Distribution

The preprocessing pipeline generated:

$$X_{train} = (23457, 53)$$

$$X_{test} = (5865, 53)$$

After preprocessing, the final feature space contained 53 numerical features which were used for regression model training.

6 Machine Learning Algorithm

Regression algorithms are supervised learning techniques used to predict continuous numerical values. In this experiment, four regression models are implemented and compared:

- Linear Regression
- Ridge Regression
- Lasso Regression
- Elastic Net Regression

6.1 Linear Regression

Linear Regression models the relationship between input features and the target variable by fitting a linear equation.

The mathematical representation is:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n$$

where:

- y represents the predicted loan sanction amount.
- $x_1, x_2, \dots, x_n$ represent input features.
- β_0 represents the intercept.
- $\beta_1, \beta_2, \dots, \beta_n$ represent feature coefficients.

The objective of Linear Regression is to minimize the residual sum of squares:

$$RSS = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

where y_i is the actual value and $\hat{y}_i$ is the predicted value.

Advantages

- Simple and easy to interpret.
- Computationally efficient.

- Works well when relationships are approximately linear.

Limitations

- Sensitive to outliers.
- Performs poorly with highly correlated features.
- Cannot capture complex nonlinear relationships.

6.2 Ridge Regression

Ridge Regression is a regularized version of Linear Regression that reduces model complexity by adding an L2 penalty term.

The objective function is:

$$Loss = \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^m \beta_j^2$$

where:

- λ is the regularization parameter.
- β_j represents model coefficients.

Ridge regression reduces large coefficient values and improves model generalization.

Advantages

- Handles multicollinearity effectively.
- Reduces overfitting.
- Maintains all features in the model.

Limitations

- Does not perform feature selection.
- Requires selection of suitable regularization parameter.

6.3 Lasso Regression

Lasso Regression applies L1 regularization by adding the absolute values of coefficients as a penalty term.

The objective function is:

$$Loss = \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^m |\beta_j|$$

The L1 penalty forces some coefficients to become exactly zero, which performs automatic feature selection.

Advantages

- Performs feature selection.
- Produces simpler models.
- Reduces overfitting.

Limitations

- May remove important features when features are correlated.
- Sensitive to regularization parameter selection.

6.4 Elastic Net Regression

Elastic Net combines both L1 and L2 regularization techniques.

The objective function is:

$$Loss = \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \lambda \left(\alpha \sum |\beta_j| + (1 - \alpha) \sum \beta_j^2 \right)$$

where:

- α controls the contribution of L1 regularization.
- $(1 - \alpha)$ controls the contribution of L2 regularization.

Elastic Net provides the benefits of both Ridge and Lasso regression.

Advantages

- Handles correlated features better than Lasso.
- Performs feature selection.
- Reduces overfitting.

Limitations

- Requires tuning of multiple hyperparameters.
- More computationally expensive.

7 Model Training

The regression models were trained using the preprocessed training dataset.

The implemented models are:

Model	Regularization	Parameters
Linear Regression	None	Default
Ridge	L2	$\alpha = 1$
Lasso	L1	$\alpha = 0.1$
Elastic Net	L1 + L2	$\alpha = 0.1$, $l1_ratio = 0.5$

Table 4: Initial Regression Model Parameters

8 Hyperparameter Tuning

GridSearchCV was used to find optimal regularization parameters.

A 5-fold cross-validation strategy was used with R^2 score as the evaluation metric.

Model	Parameter Search Space	Best Parameter
Ridge	0.01,0.1,1,10,100	$\alpha = 100$
Lasso	0.001,0.01,0.1,1,10	$\alpha = 10$
Elastic Net	α :0.01,0.1,1,10 $l1_ratio$:0.2,0.5,0.8	$\alpha = 0.1$ $l1_ratio = 0.8$

Table 5: Hyperparameter Optimization Results

The best cross-validation scores obtained were:

Model	Best CV R^2 Score
Ridge	0.5923
Lasso	0.5921
Elastic Net	0.5939

Table 6: Grid Search Cross Validation Performance

9 Experimental Setup

The experiment was implemented using Python and Scikit-Learn libraries. The following software and hardware environment was used:

Parameter	Details
Programming Language	Python
Machine Learning Library	Scikit-Learn
Data Processing Library	Pandas, NumPy
Visualization Library	Matplotlib
Development Environment	Jupyter Notebook
Python Version	Python 3.x (Anaconda Environment)
Operating System	Linux (Anaconda Environment)
Dataset Size	30000 samples
Processor	Intel based system

Table 7: Experimental Environment

Reusable Functions

The implementation was modularized using reusable functions:

- `perform_regression_eda()` for complete exploratory analysis.
- `preprocess_data()` for preprocessing pipeline generation.
- `train_regression()` for training different regression models.
- `compare_regression_models()` for calculating regression metrics.
- `tune_regression_model()` for hyperparameter optimization.
- `compare_cross_validation()` for model validation.

10 Performance Evaluation

The trained regression models were evaluated using regression performance metrics.

The following metrics were used:

10.1 Mean Absolute Error (MAE)

MAE measures the average absolute difference between actual and predicted values.

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

Lower MAE indicates better prediction accuracy.

10.2 Mean Squared Error (MSE)

MSE calculates the average squared difference between actual and predicted values.

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Lower MSE represents better model performance.

10.3 Root Mean Squared Error (RMSE)

RMSE is the square root of MSE and represents error in the same unit as the target variable.

$$RMSE = \sqrt{MSE}$$

10.4 R² Score

R² score represents the proportion of variance explained by the model.

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$$

Higher R² indicates better model fitting.

11 Test Dataset Performance

The performance of all regression models on the test dataset is shown below.

Model	MAE	MSE	RMSE	R ² Score
Linear Regression	20754.89	8.95E8	29917.03	0.6021
Ridge Regression	20768.96	8.95E8	29921.56	0.6020
Lasso Regression	20737.56	8.94E8	29903.95	0.6024
Elastic Net	20871.54	9.00E8	30000.72	0.5999

Table 8: Regression Model Performance on Test Data

Inference:

All regression models achieved similar performance because the dataset contains strong linear relationships between important features and the target variable. Lasso Regression obtained the highest R² score of 0.6024 and the lowest RMSE among the tested models. Regularization methods provided comparable performance to Linear Regression while controlling model complexity.

12 Cross Validation Performance

Five-fold cross validation was performed to verify model stability.

Model	MAE	RMSE	R ² Score
Linear Regression	21103.77	30788.17	0.5947
Ridge Regression	21108.41	30760.81	0.5954
Lasso Regression	21081.06	30750.04	0.5957
Elastic Net	21175.97	30777.02	0.5950

Table 9: 5-Fold Cross Validation Results

Inference:

The cross-validation results are consistent with the test performance. Lasso Regression achieved the highest average validation R^2 score of 0.5957, showing better generalization ability. Ridge and Elastic Net also performed consistently, indicating that regularization helped reduce model variance.

13 Training and Prediction Time Analysis

The execution time of different regression algorithms was measured.

Model	Training Time (seconds)	Prediction Time (seconds)
Linear Regression	0.0528	0.0007
Ridge Regression	0.0186	0.0017
Lasso Regression	0.4259	0.0011
Elastic Net	0.8752	0.0008

Table 10: Training and Prediction Time Comparison

Inference:

Ridge Regression required the least training time among all models. Elastic Net required the highest training time because it combines both L1 and L2 regularization. Linear Regression provided the fastest prediction time, making it computationally efficient for deployment.

14 Model Selection

Based on the evaluation metrics, Lasso Regression provided the best overall performance with:

- Lowest RMSE: 29903.95
- Highest Test R^2 Score: 0.6024
- Highest Cross Validation R^2 Score: 0.5957

Therefore, Lasso Regression was selected as the final regression model for loan sanction amount prediction.

15 Results

The performance of Linear Regression, Ridge Regression, Lasso Regression and Elastic Net Regression models were compared using the test dataset.

Table 11: Regression Model Performance Comparison

Model	MAE	MSE	RMSE	R^2
Linear Regression	20754.88	8.95×10^8	29917.03	0.6021
Ridge Regression	20768.96	8.95×10^8	29921.56	0.6020
Lasso Regression	20737.55	8.94×10^8	29903.94	0.6024
Elastic Net	20871.53	9.00×10^8	30000.72	0.5998

Inference:

Lasso Regression achieved the lowest MAE and RMSE among the evaluated models, indicating slightly better prediction accuracy. Linear Regression and Ridge Regression produced comparable performance with approximately 60% variance explained by the models. Elastic Net provided slightly lower performance due to the combined regularization effect. Overall, regularization helped control model complexity without significantly reducing predictive performance.

16 Cross Validation Results

Five-fold cross validation was performed to evaluate model stability.

Table 12: Five Fold Cross Validation Performance

Model	MAE	RMSE	R ²	Validation Error
Linear Regression	21103.76	30788.17	0.5946	9.48×10^8
Ridge Regression	21108.40	30760.80	0.5954	9.46×10^8
Lasso Regression	21081.06	30750.04	0.5956	9.45×10^8
Elastic Net	21175.96	30777.01	0.5949	9.47×10^8

Inference:

The cross-validation results show that all models produced consistent performance across different training subsets. Lasso Regression obtained the highest average R² score of 0.5956, indicating better generalization. The small difference between training and validation errors indicates that the models did not suffer from severe overfitting.

17 Hyperparameter Tuning Results

GridSearchCV was used for selecting optimal regularization parameters.

Table 13: Best Hyperparameters

Model	Parameter	Optimal Value
Ridge Regression	Alpha	100
Lasso Regression	Alpha	10
Elastic Net	Alpha	0.1
Elastic Net	L1 Ratio	0.8

Inference:

Hyperparameter tuning improved model selection by identifying suitable regularization strengths. Ridge regression selected a higher alpha value, showing that stronger L2 regularization was beneficial. Elastic Net selected a combination of L1 and L2 penalties with an L1 ratio of 0.8, providing a balance between feature selection and coefficient shrinkage.

18 Regression Visualization

18.1 Actual vs Predicted Plot

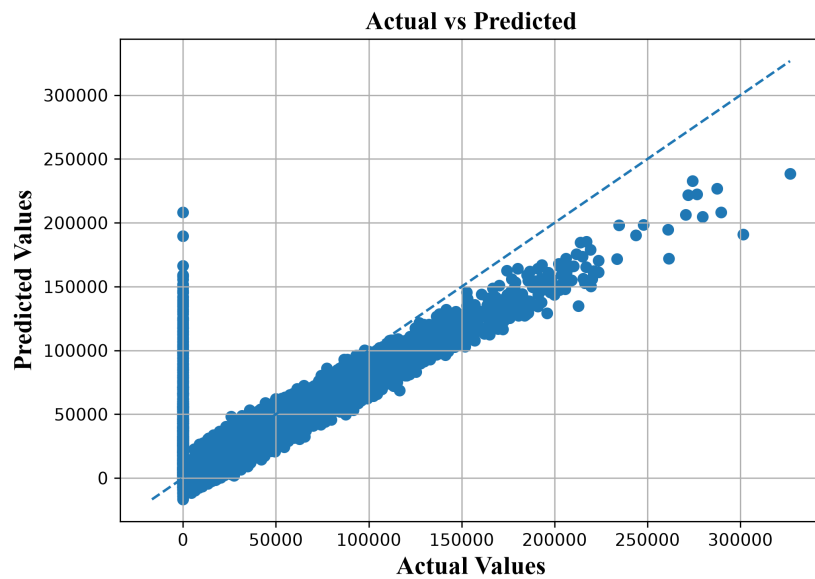


Figure 2: Actual Values vs Predicted Values

Inference:

The actual versus predicted plot shows the relationship between predicted loan sanction amounts and true values. Points closer to the diagonal line indicate accurate predictions. The model captures the overall trend of loan sanction values but some deviations are observed due to variations in customer financial characteristics.

18.2 Residual Analysis

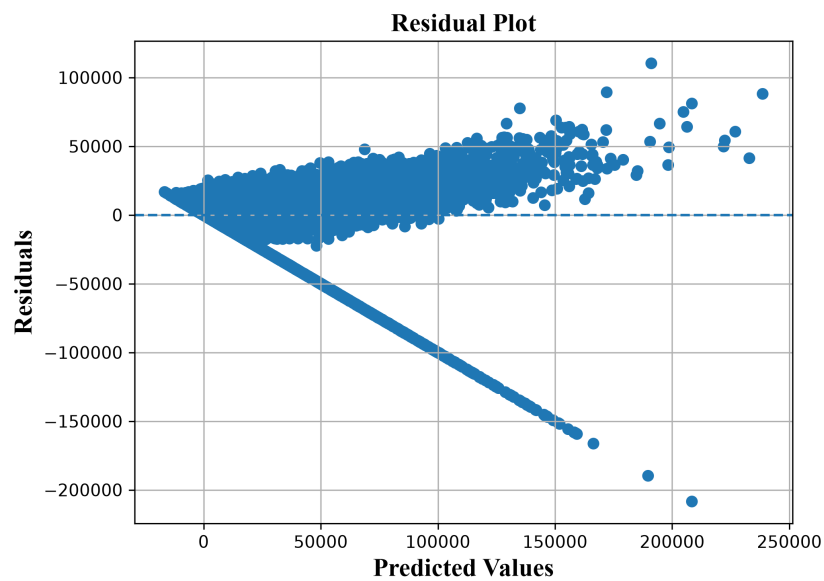


Figure 3: Residual Distribution Plot

Inference:

The residual plot is used to analyze prediction errors. Residuals are distributed around zero, indicating that the model does not show significant systematic bias. Some outliers are present, which may be due to extreme income and property values in the dataset.

19 Model Coefficient Analysis

Regression coefficients were analyzed to understand the contribution of different features.

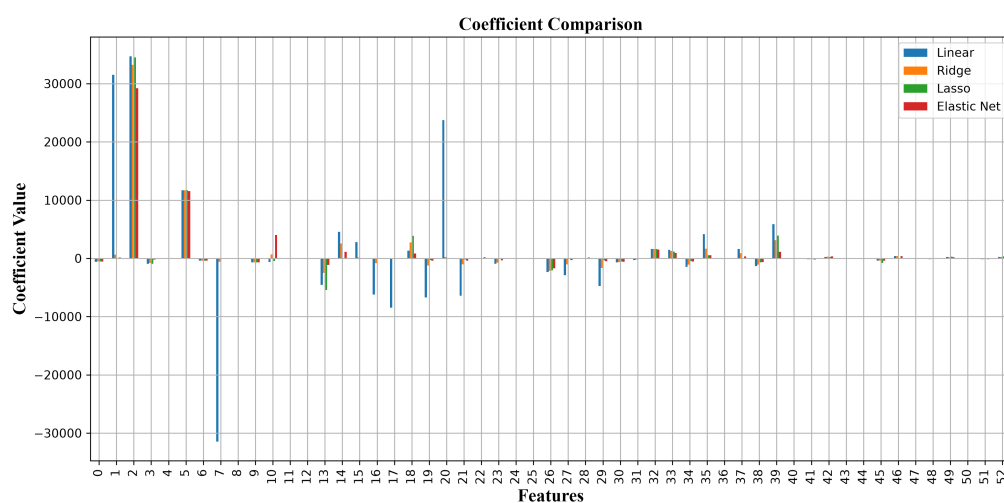


Figure 4: Coefficient Comparison Between Regression Models

Inference:

The coefficient comparison shows the effect of regularization on feature weights. Linear Regression produces larger coefficients, while Ridge, Lasso and Elastic Net reduce coefficient magnitude. Lasso performs feature selection by reducing less important feature coefficients towards zero.

20 Execution Time Analysis

Table 14: Training and Prediction Time Comparison

Model	Training Time (s)	Prediction Time (s)
Linear Regression	0.0528	0.0007
Ridge Regression	0.0186	0.0016
Lasso Regression	0.4258	0.0011
Elastic Net	0.8752	0.0007

Inference:

Linear Regression and Ridge Regression required less training time compared to Lasso and Elastic Net. Elastic Net required the highest training time due to optimization of both L1 and L2 regularization parameters. Prediction time was very small for all models, making them suitable for real-time prediction applications.

21 Discussion

The experiment implemented regression models for predicting loan sanction amounts using customer and property-related attributes.

The dataset contained numerical and categorical features with missing values. Missing values were handled using median and mode imputation, while categorical attributes were converted using one-hot encoding. Numerical features were standardized before model training.

Four regression algorithms were compared:

- Linear Regression provided a strong baseline model.
- Ridge Regression reduced coefficient magnitude using L2 regularization.
- Lasso Regression performed feature selection using L1 regularization.
- Elastic Net combined both L1 and L2 penalties.

Among all models, Lasso Regression achieved the best test performance with an R^2 score of 0.6024. The results indicate that loan sanction amount depends mainly on loan request amount, property price, current loan expenses and credit score.

22 Conclusion

This experiment successfully implemented regression analysis using Linear Regression and regularized regression techniques.

The models were trained and evaluated using multiple performance metrics including MAE, MSE, RMSE and R^2 . Hyperparameter tuning using GridSearchCV improved the selection of regularized models.

Lasso Regression achieved the best overall performance with the highest R^2 score and lowest prediction error. The experiment demonstrates that regularization techniques improve model stability and reduce unnecessary feature influence while maintaining predictive accuracy.

23 References

1. C. M. Bishop, *Pattern Recognition and Machine Learning*, Springer, 2006.
2. T. Hastie, R. Tibshirani and J. Friedman, *The Elements of Statistical Learning*, Springer, 2nd Edition, 2009.
3. S. Raschka and V. Mirjalili, *Python Machine Learning*, Packt Publishing, 3rd Edition, 2019.
4. F. Pedregosa et al., "Scikit-learn: Machine Learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.
5. Scikit-learn Documentation, <https://scikit-learn.org/>

Appendix

Implementation Details

The experiment was implemented using Python with reusable modules for:

- Data preprocessing pipeline
- Regression model training
- Hyperparameter optimization
- Cross validation
- Performance evaluation
- Visualization generation

The implementation followed modular programming practices to improve code reusability and maintainability.